



NATIONAL TEACHERS COLLEGE

Graduate School – School of Teacher Education

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Software Design & Analysis Document (SDAD) for UniQueue

by:

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May 2026

1. Process Architecture

Level-0 DFD

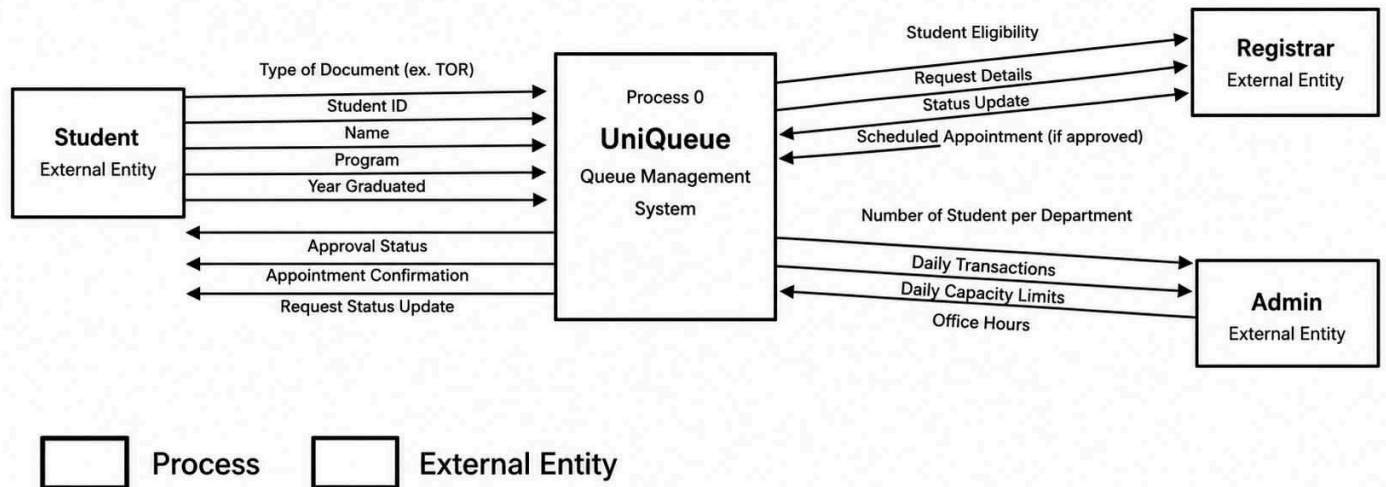


Figure 1: Level-0 Context DFD — UniQueue Queue Management System

Level-1 DFD

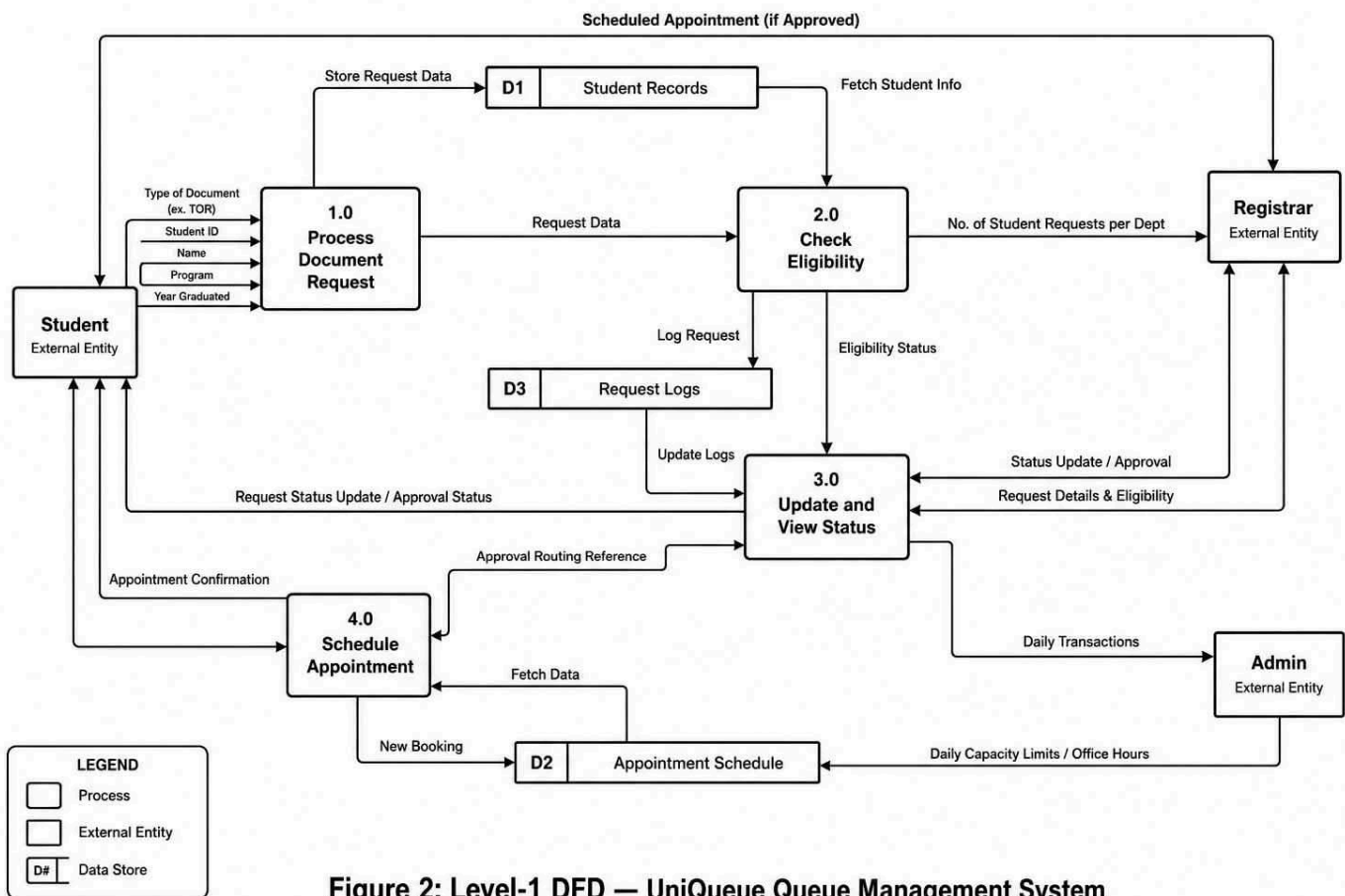
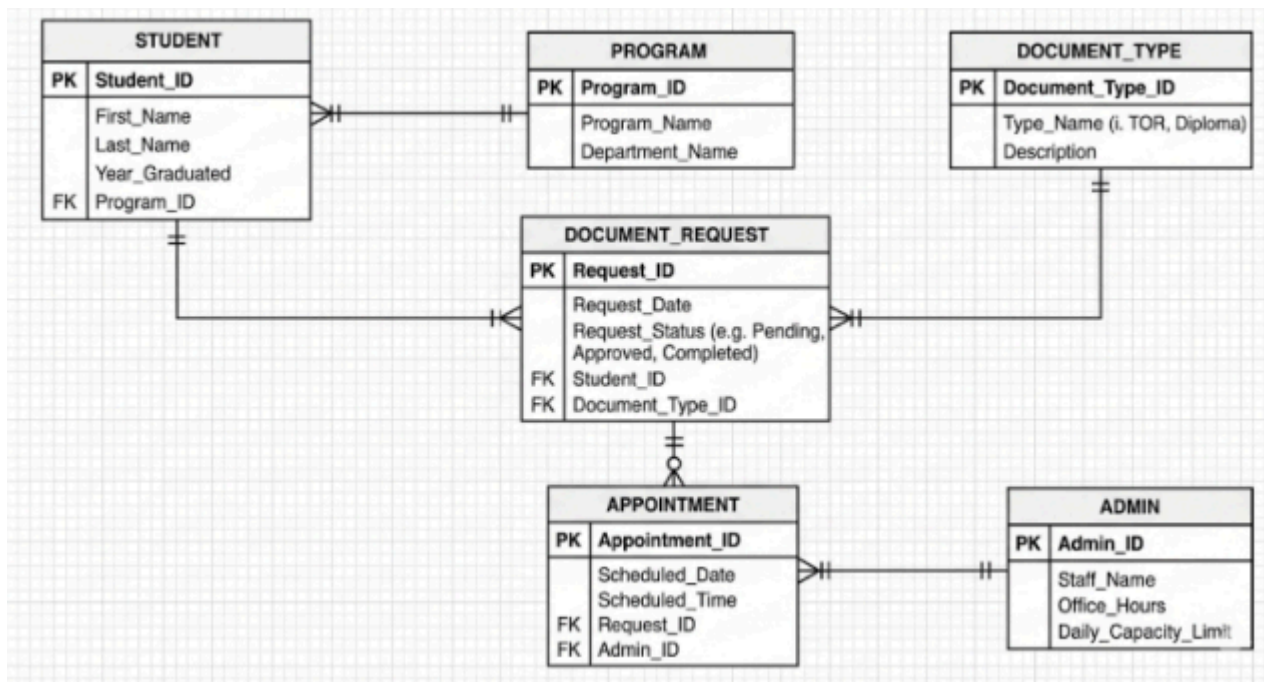
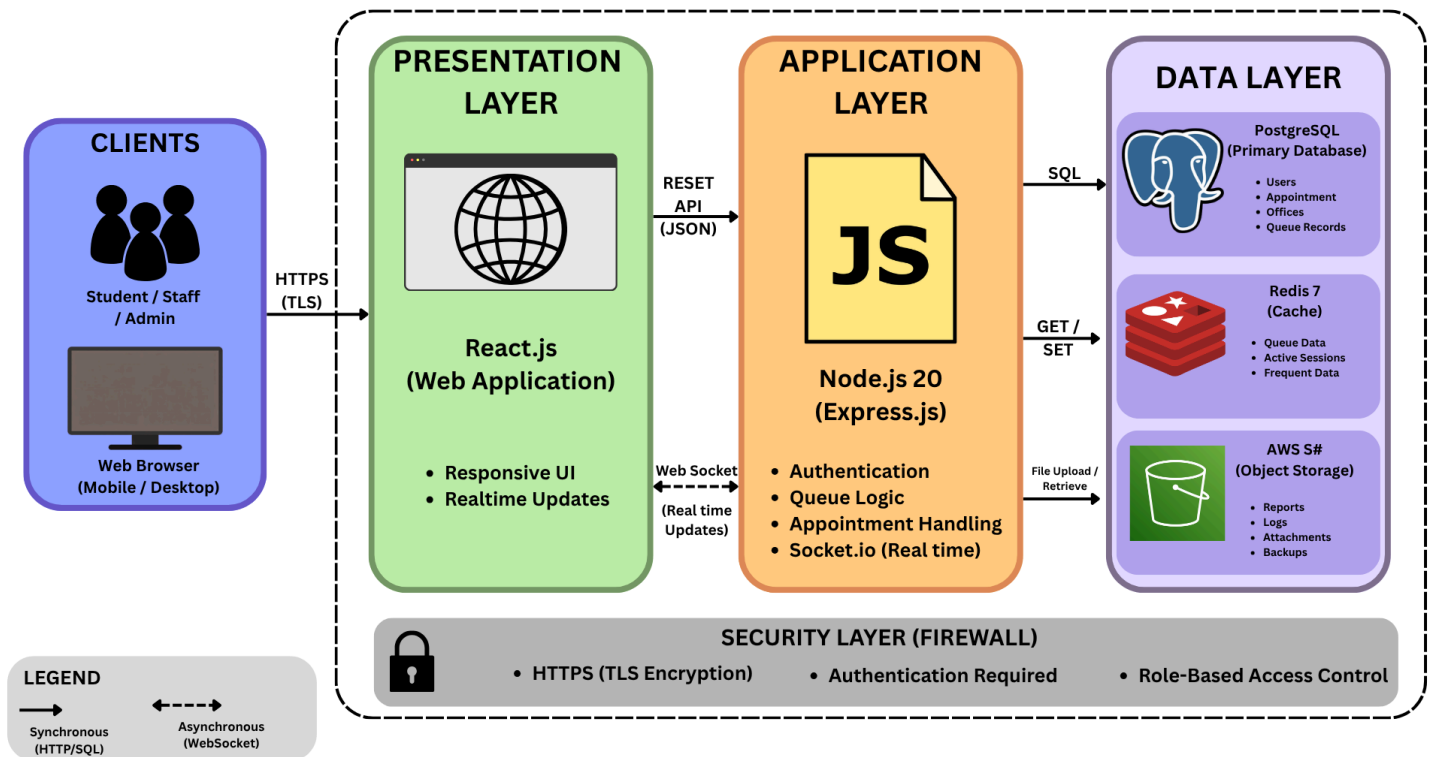


Figure 2: Level-1 DFD — UniQueue Queue Management System

2. Data Architecture



3. System Infrastructure



4. TELOS Feasibility Study

Technical	- The technical feasibility study of the UniQueue project includes determining the technical capabilities of the team to construct the proposed User Interface (UI) and an appropriate back-end database. - A prototype of the UniQueue project will be developed using web-based tools.
Economic	- The economic feasibility study of the UniQueue project will compare the tangible costs of developing the UniQueue project to the "intangible benefits" associated with the UniQueue system. Tangible Constraints: The constraints of the UniQueue project are the tools available to build and develop the project, which will be limited to the tools provided free of charge by the school, or the school itself will provide tools necessary to develop the UniQueue project.

	● Intangible Benefits: Intangible benefits of the UniQueue system will include improved student satisfaction; less stress on the school office staff; and, more students will feel like they are valued by their school.
Legal	● The UniQueue system must comply strictly with the provisions of Republic Act 10173 (The Data Privacy Act of 2012) regarding the collection, use and the management of student information.
Operational	● The operational feasibility study of the UniQueue project examines how easy it will be to integrate the digital UniQueue system into the "informal work habits" of the school staff. ● One of the major operational risks facing the UniQueue project is the potential for resistance from staff to transition from a manual logging system to using a digital UniQueue system. ● The UniQueue system would be regarded as an operational failure if it is hard for busy Registrar and Cashier Staff to operate the system during peak traffic periods.
Schedule	● The team needs to identify and achieve critical milestone deadlines for the duration of the semester so that the project is delivered on time. ●

5. UI/UX Blueprint

UQ

UNIQUEUE

UQ

UniQueue

Online Queue & Appointment Mgmt

EMAIL

your.email@university.edu

PASSWORD

.....

SIGN IN

DEMO ACCOUNTS

Student: john.doe@university.edu

Staff: maria.santos@university.edu

Admin: admin@university.edu

Password: password

Welcome, John Doe

Student ID: 2021-00001

1

ACTIVE REQUESTS

1

TOTAL REQUESTS

0

COMPLETED

QUICK ACTIONS

☐ SUBMIT NEW REQUEST

☐ VIEW MY QUEUE

RECENT REQUESTS

- **Transcript Request**
REGISTRAR Office · 2026-04-08 at 10:00

APPROVED

[← Back to Dashboard](#)

Submit New Request

Fill out the form to schedule an appointment with a university office.

SELECT OFFICE *

Choose an office



DESCRIPTION *

Provide details about your concern...

PREFERRED DATE *

dd/mm/yyyy



17

PREFERRED TIME *

--:--



CANCEL

SUBMIT REQUEST

[← Back to Dashboard](#)

Submit New Request

Fill out the form to schedule an appointment with a university office.

SELECT OFFICE *

Choose an office ▼

Registrar Office

Cashier Office

IT Office

PREFERRED DATE *

dd/mm/yyyy



CANCEL

SUBMIT REQUEST

Submit New Request

IT Office ▼

Portal Problems ▼

can't log in even with correct username and password

PREFERRED DATE *



April 2026



Su

Mo

Tu

We

Th

Fr

Sa

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

16

17

18

19

Clear

Today

PREFERRED TIME *

--:--



CANCEL

SUBMIT REQUEST

Submit New Request

IT Office ▼

Portal Problems ▼

can't log in even with correct username and password

dd/mm/yyyy

17

PREFERRED TIME *

HR

05

:

MIN

26

pm

am

06

07

08

09

10

27

28

29

30

31

CANCEL

SUBMIT REQUEST

Submit New Request

IT Office ▼

Portal Problems ▼

can't log in even with correct username and password

PREFERRED DATE *

13/04/2026



PREFERRED TIME *

03:01 pm



CANCEL

SUBMIT REQUEST

Staff Dashboard

Registrar Office

0

PENDING

0

IN
PROGRESS

0

COMPLETED

3

TOTAL

QUICK ACTIONS

☐ [MANAGE QUEUE](#)

TODAY'S APPOINTMENTS

Scheduled appointments for 4/7/2026



No appointments scheduled for today

[← Back to Dashboard](#)

My Queue

Track your appointment status in real-time

Transcript Request

REGISTRAR Office

APPROVED

REQUEST ID

REQ001

QUEUE NUMBER

#5

SCHEDULED DATE & TIME

2026-04-08 at 10:00

EST. WAIT TIME

15 minutes

Position in Queue

2 of 5

DESCRIPTION

Need official transcript for job application



[← Back to Dashboard](#)

Queue Management

Manage and process student requests

ID	STUDENT	TYPE	STATUS	ACTION	
REQ001	J. Doe	Transcript	APPROVED	Process	Done
REQ002	A. Cruz	Enrollment	PENDING	Process	
REQ003	B. Reyes	Certificate	PENDING	Process	

Staff can process, approve, or complete requests from this view.

Admin Dashboard

System overview and analytics

3

TOTAL USERS
Registered
accounts

5

TOTAL
REQUESTS
All time

1

PENDING
Needing action

1

COMPLETED
Successfully
served

QUICK ACTIONS

[MANAGE USERS](#)[SYSTEM SETTINGS](#)

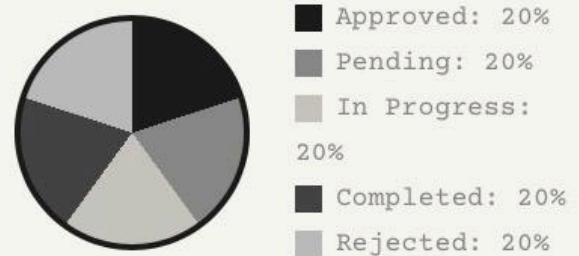
Requests by Office

Distribution across offices



Request Status Distribution

Current status of all requests



RECENT REQUESTS

Latest 5 requests across all offices

STUDENT / TYPE	DATE	STATUS
REGISTRAR John Doe Transcript Request	2026-04-07	APPROVED
CASHIER Jane Smith Payment Inquiry	2026-04-07	IN PROGRESS
IT Mike Johnson Account Access	2026-04-07	PENDING
REGISTRAR Sarah Lee Enrollment Verification	2026-04-08	COMPLETED
REGISTRAR David Brown Grade Inquiry	2026-04-12	REJECTED

[← Back to Dashboard](#)

User Management

Manage system users and their roles

[+ ADD USER](#)**1**

TOTAL STUDENTS

1

TOTAL STAFF

1

TOTAL ADMINS

ALL USERS

3 users found

**John Doe**

john.doe@university.edu

Student ID: 2021-00001

STUDENT

**Maria Santos**

maria.santos@university.edu

Office: REGISTRAR

STAFF

**Admin User**

admin@university.edu

ADMIN

[← Back to Dashboard](#)

System Settings

Configure UniQueue system parameters



Office Capacity Settings

Set daily maximum capacity for each office

REGISTRAR OFFICE DAILY CAPACITY

Maximum appointments per day: 20

CASHIER OFFICE DAILY CAPACITY

Maximum appointments per day: 15

IT OFFICE DAILY CAPACITY

Maximum appointments per day: 10



Office Hours

Set operating hours for all offices

OPENING TIME



CLOSING TIME



System Preferences

Configure system behavior and features

6. Validation Plan

Requirement ID	Requirement Description	Test Case ID	Expected Result
FR-01	Students can create accounts and log in	TC-01	Students can successfully access their accounts
FR-02	Students can generate queue tickets	TC-02	Queue tickets are generated correctly
FR-03	Students can monitor queue status in real-time	TC-03	Queue updates are displayed accurately
FR-04	Staff can manage and update queues	TC-04	Staff can modify queue records successfully
FR-05	System sends notifications to users	TC-05	Notifications are delivered properly
NFR-01	System must be user-friendly	TC-06	Users can navigate the system easily
NFR-02	System must perform during peak hours	TC-07	System remains stable under heavy usage
NFR-03	System must protect user information	TC-08	Unauthorized access is prevented
NFR-04	System must support multiple browsers and devices	TC-09	System functions correctly across platforms